

Section 6.1: Statements

Finite Mathematics MTH 132 Spring 2014

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Chapter 6: Logic

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6.1: Statements

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6.1: Statements

6.2: Truth Tables and Equivalent Statements

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6.1: Statements

6.2: Truth Tables and Equivalent Statements

6.4: Conditional Statements

Statement

A *statement* is a declarative sentence that is either true or false.

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STATEMENTS: True or
False

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- Socrates was a man. **T**
- $8 - 3 + 4 = 1$ **F**
- A triangle has three sides. **T**
- Is Socrates a man?

STATEMENTS: True or False

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- Socrates was a man. **T**
- $8 - 3 + 4 = 1$ **F**
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- Is Socrates a man?
- Go do some math.

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- $8 - 3 + 4$

STATEMENTS: True or False

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- Socrates was a man. **T**
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- A triangle has three sides. **T**

STATEMENTS: True or False

- Is Socrates a man?
- Go do some math.
- $8 - 3 + 4$
- This statement is false (*Liar's Paradox*).

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NOT STATEMENTS

Compound Statement

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- Truth or dare is a game.

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- If a polygon has three sides than go get some ice cream.
- Truth or dare is a game.
- This statement is true, and this statement is false.

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NOT COMPOUND
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Negation of Statement

- Socrates is not a man?

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- Socrates is a man.
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- $3 < 4$, "3 is less than 4"

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- $3 \geq 4$, "3 is not less than 4"

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Negation of Statement

- Socrates is not a man?
- Socrates is a man.
- $3 \geq 4$, "3 is not less than 4"
- $8 - 3 + 4 \neq 9$

Notation

	Symbol
statement	p, q, r , etc.
“and”	$\wedge$
“or”	$\vee$
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Let m represent “Socrates is a man” and r represent “it’s going to rain”.

ex. $m \wedge r$

Socrates is a man and it’s going to rain.

ex. $\sim m \vee r$

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Socrates is a man and it’s going to rain.

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Socrates if not a man or it’s going to rain.

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Let m represent “Socrates is a man” and r represent “it’s going to rain”.

ex. $m \wedge r$

Socrates is a man and it’s going to rain.

ex. $\sim m \vee r$

Socrates if not a man or it’s going to rain.

ex. $\sim (m \vee r)$

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Socrates if not a man or it’s going to rain.

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It is not case that Socrates is a man or it’s going to rain.

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It is not case that Socrates is a man or it’s going to rain.

Equivalently, $\sim m \wedge \sim r$, *Socrates is not a man and it is not going to rain.*

Your Turn: Statements

Problem 6.1.1. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

Nashville is the capital of Tennessee.

Your Turn: Statements

Problem 6.1.2. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

Got milk?

Your Turn: Statements

Problem 6.1.3. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

China does not have a population of more than 1 billion, and it is a country.

Your Turn: Statements

Problem 6.1.4. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

If I get an A then I will celebrate.

Your Turn: Negation

Problem 6.1.5. Write a negation of each statement.

This is not the time to complain.

Your Turn: Negation

Problem 6.1.6. Write a negation of each statement.

$$y \leq 5$$

Your Turn: Negation

Problem 6.1.7. Write a negation of each statement.

$$p \wedge q$$

Finding the Truth Value: Conjunction, i.e., “and”

Example 6.1.1.

Let p represent “ $5 > 3$ ” and q represent “ $6 < 0$ ”. Find the truth value of $p \wedge q$

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Let p represent “ $5 > 3$ ” and q represent “ $6 < 0$ ”. Find the truth value of $p \wedge q$

Using a *truth table*.

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Let p represent “ $5 > 3$ ” and q represent “ $6 < 0$ ”. Find the truth value of $p \wedge q$

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p	q	$p \wedge q$

- Determine which statements are true and false, and choose the corresponding answer.

Finding the Truth Value: Conjunction, i.e., “and”

Example 6.1.2.

Let p be true and q be true. Find the truth value of $p \wedge q$

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Let p be true and q be true. Find the truth value of $p \wedge q$

Using a *truth table*.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Finding the Truth Value: Conjunction, i.e., “and”

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Let p be true and q be true. Find the truth value of $p \wedge q$

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p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- Determine which statements are true and false, and choose the corresponding answer.

Finding the Truth Value: Disjunction, i.e., “or”

Example 6.1.3.

Let p represent “ $5 > 3$ ” or q represent “ $6 < 0$ ”. Find the truth value of $p \wedge q$

Finding the Truth Value: Disjunction, i.e., “or”

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Let p represent “ $5 > 3$ ” or q represent “ $6 < 0$ ”. Find the truth value of $p \wedge q$

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Finding the Truth Value: Disjunction, i.e., “or”

Example 6.1.3.

Let p represent “ $5 > 3$ ” or q represent “ $6 < 0$ ”. Find the truth value of $p \wedge q$

Using a *truth table*.

p	q	$p \vee q$
T	T	
T	F	
F	T	
F	F	

Finding the Truth Value: Disjunction, i.e., “or”

Example 6.1.3.

Let p represent “ $5 > 3$ ” or q represent “ $6 < 0$ ”. Find the truth value of $p \wedge q$

Using a *truth table*.

p	q	$p \vee q$
T	T	
T	F	
F	T	
F	F	

- Determine which statements are true and false, and choose the corresponding answer.

Compound Statements

Example 6.1.4.

Let p represent a false statement, and let q represent a true statement. Find the truth value of the compound statement

$$\sim (\sim p \wedge q)$$

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$$\sim (\sim p \wedge q)$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Compound Statements

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Let p represent a false statement, and let q represent a true statement. Find the truth value of the compound statement

$$\sim (\sim p \wedge q)$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$\sim (\sim p \wedge q)$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Compound Statements

Example 6.1.5.

Let p represent a true statement, and let q and r represent false statements. Find the truth value of the compound statement

$$(p \wedge r) \vee \sim q$$

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Let p represent a true statement, and let q and r represent false statements. Find the truth value of the compound statement

$$(p \wedge r) \vee \sim q$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Compound Statements

Example 6.1.5.

Let p represent a true statement, and let q and r represent false statements. Find the truth value of the compound statement

$$(p \wedge r) \vee \sim q$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$(p \wedge r) \vee \sim q$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Finding the Truth Value: Compound Statements

Example 6.1.6.

If q is true, what is the truth value of $q \vee (q \wedge \sim p)$

Finding the Truth Value: Compound Statements

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If q is true, what is the truth value of $q \vee (q \wedge \sim p)$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
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Finding the Truth Value: Compound Statements

Example 6.1.6.

If q is true, what is the truth value of $q \vee (q \wedge \sim p)$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$$q \vee (q \wedge \sim p)$$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Your Turn: Find the Truth Value

Problem 6.1.8. Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

$$\sim p$$

Your Turn: Find the Truth Value

Problem 6.1.9. Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

$$p \vee \sim q$$

Your Turn: Find the Truth Value

Problem 6.1.10. Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

$$\sim (p \vee \sim q)$$

Your Turn: Find the Truth Value

Problem 6.1.11. Let p represent a true statement, and q and r represent false statements. Find the truth value of each compound statement.

$$(q \vee \sim r) \wedge p$$

Your Turn: Find the Truth Value

Problem 6.1.12. Let p represent a true statement, and q and r represent false statements. Find the truth value of each compound statement.

$$(\sim p \wedge q) \vee \sim r$$

Your Turn: Find the Truth Value

Problem 6.1.13. Let p be the statement " $2 > 7$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

$$\sim p \vee \sim r$$

Your Turn: Find the Truth Value

Problem 6.1.14. Let p be the statement " $2 > 7$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

$$(p \wedge q) \vee r$$

Your Turn: Find the Truth Value

Problem 6.1.15. Let p be the statement " $2 > 7$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

$$(\sim r \wedge q) \vee \sim p$$

Answers

Answers: T, F, T, T, T, F, T, T